

CineStats

Global Box Office & TV Analytics Tracker

Project Requirements & Technical Specification — v2.0

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Document Type	Project Requirements & Technical Specification
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Platform	Streamlit Web Application
Monetization	Free Forever — No Ads, No Paid Tier
Default Theme	Dark Mode (Light Mode available)
Status	Planning / Pre-Development

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1 Executive Summary

CineStats is a modern, sleek, dark-mode-first interactive web dashboard that aggregates and visualizes box office and viewership data for films, film series, cinematic franchises, and TV series. Built on a Python + Streamlit stack, it is permanently free with no advertising and no paid tier.

The application is structured around four primary content types: individual Movies, Series & Franchises (multi-film sagas and cinematic universes), TV Series (tracked as a fully separate module), and Comparisons (up to 10 entities side-by-side). Each has its own dedicated tab and detail view in the sidebar.

CineStats delivers immediate value on first load: the Home tab populates a full, rich dashboard with no user input. Data is cached per session to avoid redundant scraping. Users are free to explore any movie, series, or franchise they like; they simply cannot re-fetch the exact same data set repeatedly within the same session.

Every view in CineStats supports one-click PDF report export — a fully formatted, visualization-rich document the user can download and save.

2 Project Overview

2.1 Goals & Objectives

- Deliver a fully-populated Home tab on first load with no required user interaction.
- Present a unified Global + India perspective serving Indian audiences alongside international users.
- Support Movie Detail views, Series/Franchise views (all entries, no internal limit), and TV Series views as separate first-class sections.
- Allow cross-entity comparison of up to 10 movies, series, or franchises in the Compare tab. TV-to-TV comparison is separate; TV cannot be mixed with films.
- Enforce session-level data caching to prevent scrape spam on the same view, while allowing free exploration of different entities.
- Provide downloadable, visualization-rich PDF reports for any view.
- Support multi-currency display with a persistent currency selector.
- Be permanently free — no ads, no subscriptions, no paywalls, no data selling.

2.2 Target Audience

- Indian film enthusiasts tracking both Hollywood and Bollywood/regional releases.

- Box office analysts and hobbyist researchers seeking comparative and historical data.
- TV viewers interested in ratings, viewership trends, and season-level breakdowns.
- Developers and students exploring Streamlit dashboard and scraping patterns.

2.3 Scope & Boundaries

In Scope	Film box office data (global + India), TV series ratings/viewership, series & franchise aggregation, comparison tool (up to 10), PDF export, multi-currency display, dark/light mode, sidebar navigation.
Out of Scope	Ticket booking, OTT/streaming data (future), social media integration, user accounts, push notifications to external services.
TV vs Film Rule	TV series is a fully separate module and tab. TV series cannot be compared with films or film franchises. TV-to-TV comparison is supported.
Within a Series	There is no limit on the number of films displayed within a single series or franchise detail view. All entries are shown.
Comparison Limit	Up to 10 entities (movies, series, or franchises) in the Compare tab simultaneously.
Monetization	Free forever. No advertising. No premium tier.
Deployment	Streamlit Community Cloud (free tier). SQLite for local persistence.
Legal	Personal and educational use only. No commercial redistribution of scraped data.

3 Navigation & UI Structure

3.1 Sidebar Navigation

CineStats uses a persistent left sidebar as its primary navigation. The sidebar is always visible across all views and contains the following tabs:

Home	Default landing view. Fully populated on first load. Shows Top of Year (Global + India), Releases Today/This Week, and Now Playing.
Search	Unified search across all movies, series, franchises, and TV series. Navigates to individual detail views on selection.
Compare	Compare up to 10 movies, series, or franchises side-by-side with tables and overlaid charts. TV comparison is a separate sub-section here.
TV Series	Dedicated module for television series. Fully separate from the film sections.
Settings	Currency selector, dark/light mode toggle, and session preferences.

Clicking any movie, series, or franchise card in Home or Search navigates to that entity's detail view, rendered within the main content area. A breadcrumb trail shows the navigation path.

3.2 Dark & Light Mode

- Default is Dark Mode — dark backgrounds, high-contrast text, vivid accent colours. This is the primary design target and should look polished and premium.
- Light Mode is available via a toggle in Settings or a quick-access icon in the top-right corner. Light mode is colourful, modern, and sleek — not a plain white page. It uses vibrant accent colours, soft card shadows, and a carefully chosen palette to remain visually rich.
- Mode preference persists for the session via Streamlit session state.
- All Plotly charts use matching dark/light templates (plotly_dark vs a custom light template).
- Exported PDFs use a neutral light theme regardless of the active UI mode, ensuring readability when printed.

3.3 Currency Selector

Users choose a preferred display currency from the Settings panel. This affects all monetary figures across the app.

Supported Currencies	USD (\$), INR (₹), EUR (€), GBP (£), JPY (¥), AED (إ.د.), AUD (A\$), CAD (C\$), SGD (S\$)
Default	INR (₹) — primary audience is Indian.
Worldwide Gross Rule	Worldwide and International figures always show USD as the primary figure. If the selected currency is not USD, the converted amount appears in brackets: e.g. \$599M (₹4,980 Cr). If USD is selected, only USD is shown.
India Figures	India Net and India Gross always show INR (₹ Cr) as the primary figure. Converted value shown in brackets if selected currency differs.
Exchange Rates	Fetches once per session from a free public exchange rate API (e.g. exchangerate.host or open.er-api.com) and cached in session state. Conversion is informational only — all authoritative figures remain in their native currency.
Rate Staleness	A visible 'Rates as of [time]' label is shown wherever converted figures appear, so users understand they are indicative.

4 Home View (Default Tab)

The Home tab is the default landing page. It is fully populated on first load with no user input. It is organised into three clearly labelled, visually separated sections. The Home tab is read-only — users navigate to Search or Compare for interactive exploration.

4.1 Top Movies of the Year

Displayed as two side-by-side columns or switchable sub-tabs:

Global Top (from Box Office Mojo)

- Worldwide gross leaders for the current calendar year.
- Data columns: Rank, Title, Worldwide Gross (USD + brackets if non-USD selected), Domestic %, Foreign %, Days in Release, Verdict.
- Visualisation: Interactive Plotly horizontal bar chart — Top 10 Worldwide Grosses.
- Visualisation: Plotly pie chart — Domestic vs Foreign split for the highlighted film.

Indian Top (from Sacnilk)

- India Net leaders covering Bollywood, Tamil, Telugu, Malayalam, Kannada, and other regional industries.
- Data columns: Rank, Title, Language, India Net (₹ Cr), India Gross (₹ Cr), Verdict, Days in Release.
- Visualisation: Interactive Plotly horizontal bar chart — Top 10 India Net.
- Clicking any film in either section navigates to its Movie Detail View.

4.2 Releases Today / This Week

International Releases

- Hollywood and global films released this week, plus notable historical releases on today's date.
- Shows opening day/weekend performance and current gross if still running.

Indian Releases

- Bollywood and regional Indian films released this week, plus historical releases on today's date.
- Shows opening day India Net/Gross and verdict where available.

4.3 Now Playing (Currently in Theaters)

- Films currently running in cinemas worldwide and in India.
- Displays: daily gross, total gross, holdover %, days in release, theater count where available.
- Quick filter toggle between Global focus and India focus.

- Sortable by current gross, days in release, or weekly change.
- Clicking any entry navigates to its Movie Detail View.

5 Movie Detail View

Accessed by clicking any film card in Home, Search results, or within a Series/Franchise view. Displays comprehensive performance data for a single film.

Header	Title, release date, language, genre tags (from TMDB future scope), poster thumbnail (future scope), verdict badge.
Key Metrics Bar	Worldwide Gross, Domestic Gross, India Net, India Gross, Days in Release, Theater Count — displayed as prominent stat cards.
Regional Breakdown	Expandable table: US, India, China, UK, Overseas breakdown. India row further splits into Hindi vs regional language collections where available.
Daily/Weekly Chart	Plotly line chart of daily gross over the full run (global and India on same chart with dual Y-axes if needed).
Cumulative Chart	Plotly cumulative gross trajectory chart.
Opening vs Lifetime	Opening weekend gross, opening weekend % of total, and a projected lifetime gross estimate with confidence range.
On This Day	Historical spotlight: other films that opened or peaked on the same calendar date in prior years, with links to compare.
Verdict Context	How the film's verdict compares to others in its language, genre, and franchise (if applicable).
Series Link	If the film belongs to a series or franchise, a prominent link to that Series/Franchise view.
PDF Export	One-click button to export the full detail view as a PDF report.
Compare Button	Add this film to the Compare tab (up to 10 total).

6 Series & Franchise View

Accessed by clicking a series/franchise card in Search, or via a film's 'Series Link'. Displays aggregated and per-entry data for all films in the series. There is no limit on the number of films displayed — every entry in the series is shown.

Header	Series/Franchise name, total number of films, cumulative worldwide gross, cumulative India Net, franchise lifespan (first to latest release).
Cumulative Stat Cards	Total Worldwide Gross, Total India Net, Average Film Gross, Highest-

	Grossing Entry, Lowest-Grossing Entry.
All Films Table	Full table listing every film in the series: Rank (within series), Title, Year, Worldwide Gross, India Net, Verdict, Days in Release.
Per-Film Bar Chart	Plotly grouped bar chart: Worldwide Gross and India Net per film, sorted chronologically.
Trajectory Chart	Plotly line chart showing how cumulative franchise gross grew across each new release.
Verdict Distribution	Plotly pie/donut chart showing verdict breakdown across all films in the franchise.
Regional Split	Stacked bar chart: Domestic vs International split per film.
Individual Film Links	Each film row links to its own Movie Detail View.
PDF Export	One-click full series report export as PDF.
Compare Button	Add this series to the Compare tab (counts as 1 entity toward the 10-entity limit).

7 TV Series Module

TV Series is a fully independent module accessible from the dedicated TV Series tab in the sidebar. TV data is sourced and stored separately from film data. TV series cannot be compared with films or film franchises.

7.1 TV Series List View

- Browsable list of tracked TV series with search and filter (network, genre, status: Ongoing/Ended, country of origin).
- Shows: Series Name, Network/Streamer, Seasons, Episode Count, Status, Average Rating/Viewership.

7.2 TV Series Detail View

Header	Series name, network, genre, premiere date, status (Ongoing / Ended), country.
Season Overview	Table listing all seasons: Season number, episode count, premiere date, average viewership, average rating, peak episode viewership.
Viewership Chart	Plotly line chart of episode-by-episode viewership across all seasons (with season markers).
Rating Chart	Plotly line chart of episode ratings (IMDb, Rotten Tomatoes audience score, or equivalent) across all seasons.

Season Comparison	Bar chart: average viewership per season.
Episode Table	Expandable per-season episode list with title, air date, viewership, and rating.
PDF Export	Full TV series report export as PDF.
Compare Button	Add to TV Compare (separate from film compare, max 10 TV series).

7.3 TV Series Data Sources

- Primary: TVMaze API (free, no key required) for episode listings, air dates, and network data.
- Ratings: IMDb ratings via scraping or a community dataset where available.
- US Network Viewership: Wikipedia episode tables (scraped) for major US broadcast/cable networks.
- Note: Comprehensive viewership data may not be available for all series. The UI handles missing data gracefully with 'Data unavailable' labels rather than errors.

8 Search

The Search tab provides a unified search experience across all content types in CineStats.

Search Bar	A prominent search input at the top of the tab. Searches as the user types (debounced at 300ms) against the local SQLite database.
Content Types	Results are grouped into: Films, Series & Franchises, and TV Series — each in its own clearly labelled results section.
Filters	Year range, language, region (Global / India), verdict, minimum worldwide gross, minimum India Net, content type toggle (Film / Series / TV).
Result Cards	Each result shows: title, year, type badge (Film / Series / TV), key metric (worldwide gross or India Net), verdict badge. Clicking navigates to the relevant detail view.
No Results Handling	If no local results found, CineStats offers to perform a live scrape lookup. This is the only user-triggered scrape outside the session-initial data load.
Recent Searches	Last 10 search queries stored in session state and shown as quick-access chips below the search bar.

9 Compare Tab

The Compare tab allows side-by-side analysis of multiple entities. Two separate comparison contexts exist: Film/Franchise comparisons and TV Series comparisons. These are never mixed.

9.1 Film / Franchise Comparison

Entity Limit	Up to 10 movies, series, or franchises simultaneously. Mix of individual films and series/franchises is allowed.
Selection	Entities are added via the search-and-add input in the Compare tab, or via the 'Add to Compare' button on any Movie or Series detail view.
Metrics Table	Side-by-side table: Worldwide Gross, Domestic Gross, Foreign Gross, India Net, India Gross, Opening Weekend, Days in Release, Verdict — one column per entity.
Bar Chart	Grouped Plotly bar chart comparing worldwide gross and India Net across all selected entities.
Trajectory Overlay	Overlaid Plotly line chart showing cumulative gross over time for each entity, useful for comparing opening trajectories.
Verdict Matrix	Visual grid showing verdict for each entity, colour-coded by outcome.
Regional Pie	Per-entity small pie charts showing domestic vs international split, arranged in a grid.
Clear / Remove	Users can remove individual entities from the comparison or clear all and start fresh.

9.2 TV Series Comparison

Entity Limit	Up to 10 TV series simultaneously.
Separation	Completely separate from Film comparison. Rendered as a distinct sub-section within the Compare tab.
Metrics Table	Side-by-side: Total Seasons, Total Episodes, Average Viewership, Peak Viewership, Average Rating, Status.
Viewership Chart	Overlaid line charts of average viewership per season, one line per series.
Rating Chart	Overlaid line charts of average rating per season, one line per series.
Clear / Remove	Individual removal or full clear.

10 PDF Report Export

Every detail view and the Compare view includes a prominent 'Download PDF Report' button. The exported PDF is fully formatted, includes data tables and chart images, and is suitable for saving or sharing.

Library	ReportLab (Python) for PDF generation. Plotly charts are exported as static PNG images via kaleido and embedded in the PDF.
Cover Page	Report type (Movie / Series / Franchise / Comparison / TV Series), entity name(s), date generated, CineStats branding.
Data Tables	All tables visible in the UI are reproduced in the PDF using ReportLab TableStyle with colour-coded headers matching the app's palette.
Charts	All Plotly visualisations are rendered to PNG at 150dpi and embedded at appropriate sizes. Chart titles and axis labels are preserved.
Currency Note	A footer note states the currency used for converted figures and the exchange rate at time of export.
Theme	PDFs always use a clean, light neutral theme regardless of the active UI mode, ensuring legibility when printed.
File Naming	Auto-named: CineStats_[EntityName]_Report_[YYYY-MM-DD].pdf
Scope	Available from: Movie Detail, Series/Franchise Detail, TV Series Detail, and Compare views.

11 Core Features Reference

Feature	Description
Search & Filters	Unified search across films, series, franchises, TV series. Filters by year, language, region, verdict, gross threshold, content type.
Movie Detail View	Full single-film breakdown: regional data, daily/cumulative charts, opening vs lifetime, On This Day, verdict context.
Series & Franchise View	All films in a series with no internal entry limit. Aggregated stats, per-film bar/line charts, trajectory, verdict distribution.
TV Series Module	Separate tab: series list, detail view with season/episode data, viewership and rating charts. Cannot be mixed with film comparisons.
Compare (Films)	Up to 10 films/series/franchises side-by-side with metrics table, bar charts, trajectory overlay, verdict matrix.
Compare (TV)	Up to 10 TV series side-by-side with viewership and rating overlays. Separate from film comparison.
Daily / Weekend Movers	Home tab section: biggest daily gainers and losers with % change. Refreshed on session load.
Opening Weekend Predictor	Historical-average model projecting lifetime gross from opening weekend with a confidence range indicator.
Clash Analyzer	Special view for same-release-date films showing side-by-side performance and verdict outcomes.
Regional Breakdown	US, India, China, UK, Overseas splits. India row further shows Hindi vs

Feature	Description
	regional language collections.
On This Day Spotlight	Historical section on Movie Detail: top earners from the same calendar date in prior years, linkable to Compare.
Currency Selector	Persistent per-session currency choice. Worldwide/International always shows USD; converted value in brackets if different currency selected.
Dark / Light Mode	Dark mode by default (premium, sleek). Light mode is colourful and modern. Toggle in sidebar Settings or top-bar icon.
PDF Report Export	One-click download of fully formatted PDF with data tables and embedded chart images. Available on all detail and compare views.
Session Cache	Data fetched once per session per entity. Users cannot spam-refresh the same view. Exploring different entities always works.
Genre & Studio Aggregates	Roll-up views of top films by genre or studio with Plotly charts.
Watchlist	Per-session watchlist. Add films/series to track. Optional milestone display (e.g. crossed ₹100 Cr).

12 Data Sources

12.1 Box Office Mojo (Global Film Data)

- Worldwide gross totals (current year and historical).
- Domestic (US/Canada) and International splits.
- Daily and weekly performance charts.
- Currently In Theaters listing with daily and cumulative grosses.
- Opening weekend data and theater counts.

12.2 Sacnilk (India Film Data)

- India Net and India Gross in INR Crores for Bollywood, Tamil, Telugu, Malayalam, Kannada, and other regional industries.
- Day-wise India collections for recent releases.
- Release dates and official verdicts.
- Currently running Indian films with daily collections.

12.3 TV Data Sources

- TVMaze API — free, no API key required. Provides episode listings, air dates, network, genre, and series status.

- Wikipedia episode tables — scraped for US network/cable viewership figures where available.
- IMDb ratings — scraped or sourced from community datasets for episode and series ratings.

12.4 Data Matching & Unification

- Title normalization: lowercased, punctuation stripped, whitespace normalized before matching.
- Fuzzy string matching (rapidfuzz library) to handle minor title discrepancies between sources.
- Release date proximity matching within a configurable window (default ± 7 days) to handle India vs global release gaps.
- Manual override mapping table in SQLite for edge cases such as dubbed title variations.
- A match confidence score is computed and stored; uncertain matches are flagged visually in the UI.

13 Technology Stack

Layer	Technology	Purpose
Language	Python 3.11+	Core application language.
Web Scraping	requests + BeautifulSoup4	HTTP requests and HTML parsing for BOM and Sacnilk.
Tabular Parsing	pandas.read_html	Structured HTML table extraction.
Fuzzy Matching	rapidfuzz	Title matching between data sources.
Data Processing	pandas	Cleaning, normalization, calculations, cross-source merging.
TV Data	TVMaze REST API	Episode, season, and series metadata for TV module.
Exchange Rates	exchangerate.host or open.er-api.com	Free currency rate API, fetched once per session.
Database	SQLite3 (sqlite3 stdlib)	Local persistence: movies, series, TV, daily data, watchlist, match overrides, cache metadata.
UI Framework	Streamlit	Interactive web dashboard with sidebar, tabs, session state, theming.
Visualisations	Plotly (Express + Graph Objects)	Interactive bar, line, pie, donut, and stacked charts. Dark and light templates.
PDF Generation	ReportLab	PDF report creation with styled tables and embedded images.

Layer	Technology	Purpose
Chart Export	kaleido	Static PNG export of Plotly charts for embedding in PDFs.
Deployment	Streamlit Community Cloud	Free-tier hosting via GitHub integration.
Dependency Mgmt	pip + requirements.txt	Package management and reproducible environment.

14 System Architecture

14.1 Data Layer

- **Scraper Modules:** One module per source (`bom_scraper.py`, `sacnilk_scraper.py`, `tvmaze_client.py`, `wikipedia_tv_scraper.py`). Each returns a clean, typed pandas DataFrame.
- **Cache Registry:** SQLite table `scrape_cache` stores (`source`, `entity_key`, `scraped_at`) to track what has been fetched in the current session.
- **Upsert Logic:** All writes use `INSERT OR REPLACE`. Schema migrations handled by a versioned `init_db.py` script.
- **Exchange Rate Store:** Session-level dict in Streamlit `session_state`. Not persisted to SQLite.

14.2 Business Logic Layer

- **TitleMatcher:** Fuzzy-matching engine producing unified movie objects from BOM + Sacnilk records.
- **FranchiseGrouper:** Auto-groups films into series using title pattern matching and a manual override table.
- **PredictorEngine:** Historical-average model for opening weekend to lifetime gross projection.
- **ClashDetector:** Groups films with the same release date for Clash Analyzer view.
- **MoverCalculator:** Computes daily gross deltas for the Home movers section.
- **CurrencyConverter:** Applies the session exchange rate to produce display strings in the selected currency.
- **PDFReportBuilder:** Orchestrates ReportLab layout and kaleido chart image generation for export.

14.3 Presentation Layer

- `app.py`: Main Streamlit entry point. Defines sidebar navigation and routes to page modules.
- `pages/`: One module per tab (`home.py`, `search.py`, `compare.py`, `tv_series.py`, `settings.py`).
- `components/`: Reusable UI components (`movie_card.py`, `stat_bar.py`, `chart_wrapper.py`, `pdf_button.py`).

- `theme.py`: Central colour and style constants shared across all page and component modules.

14.4 Data Flow

1. On first load of a tab or entity, Streamlit checks the `scrape_cache` table for a matching session entry.
2. If no session entry exists, the relevant scraper is called. Rate-limiting delays are applied.
3. Scraped data is cleaned, matched, and upserted into SQLite. The cache entry is written.
4. Subsequent loads of the same view within the session read directly from SQLite — no re-scrape.
5. The user may freely navigate to different movies, series, or franchises; each triggers its own cache check.
6. User interactions (search, filter, currency change, compare add/remove) operate on in-memory DataFrames loaded from SQLite — no network calls.
7. PDF export reads the current in-memory state, renders Plotly charts to PNG via kaleido, and builds the report with ReportLab — all locally.

15 Database Schema

All tables in `cineStats.db` use SQLite3. Schema versioning is handled by a `db_version` table and a migration runner in `init_db.py`.

movies

id	INTEGER PRIMARY KEY AUTOINCREMENT
title	TEXT NOT NULL — normalized title used as matching key
title_display	TEXT — original display title
release_date	DATE — YYYY-MM-DD
language	TEXT — primary language
franchise_id	INTEGER REFERENCES franchises(id) — nullable
worldwide_gross_usd	REAL
domestic_gross_usd	REAL
foreign_gross_usd	REAL
india_net_cr	REAL — India Net in INR Crores
india_gross_cr	REAL
opening_weekend_usd	REAL
theater_count	INTEGER

verdict	TEXT
days_in_release	INTEGER
source	TEXT — 'bom', 'sacnilk', or 'merged'
match_confidence	REAL — 0.0 to 1.0; null if single-source
last_updated	TIMESTAMP

franchises

id	INTEGER PRIMARY KEY AUTOINCREMENT
name	TEXT NOT NULL
name_normalized	TEXT — for matching
total_films	INTEGER
first_release	DATE
latest_release	DATE
cumulative_worldwide	REAL — USD
cumulative_india_net	REAL — INR Cr
last_updated	TIMESTAMP

daily_performance

id	INTEGER PRIMARY KEY AUTOINCREMENT
movie_id	INTEGER REFERENCES movies(id)
date	DATE
daily_gross_usd	REAL
daily_india_net_cr	REAL
cumulative_gross_usd	REAL
cumulative_india_net	REAL
theater_count	INTEGER

tv_series

id	INTEGER PRIMARY KEY AUTOINCREMENT
name	TEXT NOT NULL
network	TEXT
genre	TEXT

status	TEXT — 'Ongoing' or 'Ended'
premiere_date	DATE
total_seasons	INTEGER
total_episodes	INTEGER
avg_rating	REAL
tmvaze_id	INTEGER
last_updated	TIMESTAMP

tv_episodes

id	INTEGER PRIMARY KEY AUTOINCREMENT
series_id	INTEGER REFERENCES tv_series(id)
season	INTEGER
episode	INTEGER
title	TEXT
air_date	DATE
viewership_millions	REAL — nullable; US viewership where available
rating	REAL — IMDb or equivalent

scrape_cache

id	INTEGER PRIMARY KEY AUTOINCREMENT
source	TEXT — 'bom', 'sacnilk', 'tmvaze', 'wikipedia_tv'
entity_key	TEXT — normalized identifier for what was scraped
session_id	TEXT — Streamlit session ID
scraped_at	TIMESTAMP

watchlist

id	INTEGER PRIMARY KEY AUTOINCREMENT
entity_type	TEXT — 'movie', 'franchise', 'tv_series'
entity_id	INTEGER
alert_usd_milestone	REAL — nullable
alert_inr_milestone	REAL — nullable
added_at	TIMESTAMP

16 Data Refresh & Caching Policy

Box office data is not updated continuously on source sites — figures typically update once daily or less. CineStats therefore applies a session-based caching policy rather than a time-based one, balancing data freshness with responsible scraping behaviour.

Session Cache	Each scraped entity is cached for the duration of the user session (identified by Streamlit session ID). The same entity will not be re-scraped within the same session under any circumstance.
No Spam Refresh	Users cannot manually trigger a refresh of the same view/entity within a session. There is no refresh button on data views.
New Entity = New Fetch	Navigating to a different movie, series, or franchise always works freely. Each new entity gets its own cache entry and fetches fresh data if not already in SQLite.
Between Sessions	On a new session start, all prior session cache entries are treated as expired. Fresh scrapes occur for any entities loaded in the new session.
SQLite Persistence	Data fetched in prior sessions remains in SQLite and serves as a fallback if a live scrape fails. The UI shows a 'Last updated: [date]' label when serving persisted data.
Exchange Rates	Fetched once per session at startup and stored in session state. Not persisted to SQLite.

17 Functional Requirements

The following table captures formal functional requirements using SHALL notation. Each requirement has a unique ID for traceability.

ID	Requirement	Priority
FR-01	The system SHALL display a fully populated Home tab on first load without any user input.	Must Have
FR-02	The system SHALL display global top grossers (Worldwide Gross in USD) from Box Office Mojo.	Must Have
FR-03	The system SHALL display Indian top grossers (India Net in INR Cr) from Sacnilk.	Must Have
FR-04	The system SHALL display a Now Playing section for both global and India films.	Must Have
FR-05	The system SHALL provide a unified search across films, series, franchises, and TV series.	Must Have

ID	Requirement	Priority
FR-06	The system SHALL display a Movie Detail View with regional breakdown, daily charts, and cumulative charts.	Must Have
FR-07	The system SHALL display a Series/Franchise View with all films in the series (no entry limit).	Must Have
FR-08	The system SHALL provide a TV Series module as a separate tab, with series list and detail views.	Must Have
FR-09	TV series SHALL NOT be comparable with films or film franchises.	Must Have
FR-10	The system SHALL allow comparison of up to 10 films, series, or franchises simultaneously.	Must Have
FR-11	The system SHALL allow comparison of up to 10 TV series simultaneously (separate from film compare).	Must Have
FR-12	The system SHALL NOT re-scrape the same entity within the same session.	Must Have
FR-13	The system SHALL allow users to freely navigate to any different movie, series, or franchise.	Must Have
FR-14	The system SHALL provide a one-click PDF report export on all detail and compare views.	Must Have
FR-15	Exported PDFs SHALL include all data tables and embedded chart images from the current view.	Must Have
FR-16	The system SHALL support a currency selector with at least USD, INR, EUR, GBP, JPY, AED.	Must Have
FR-17	Worldwide/International gross SHALL always display USD as the primary figure.	Must Have
FR-18	If a non-USD currency is selected, converted amounts SHALL appear in brackets next to USD figures.	Must Have
FR-19	The system SHALL default to Dark Mode on first load.	Must Have
FR-20	The system SHALL provide a Light Mode toggle accessible from the sidebar Settings.	Must Have
FR-21	The system SHALL provide a sidebar with Home, Search, Compare, TV Series, and Settings tabs.	Must Have
FR-22	The system SHALL show an Opening Weekend Predictor with confidence range on Movie Detail.	Should Have
FR-23	The system SHALL show an On This Day section on Movie Detail linking to comparable historical films.	Should Have
FR-24	The system SHALL show a Clash Analyzer for films sharing the same release date.	Should Have
FR-25	The system SHALL support a per-session Watchlist.	Should Have
FR-26	The system SHALL show Daily/Weekend Movers on the Home tab.	Should Have

ID	Requirement	Priority
FR-27	The system SHALL show Genre and Studio aggregate views.	Could Have
FR-28	The system SHALL allow CSV export of any data table.	Could Have

18 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	The Home tab SHALL complete its initial render (from cached SQLite data) within 3 seconds on a standard broadband connection.
NFR-02	Performance	Live scrape operations SHALL complete within 15 seconds per source. A loading spinner SHALL be displayed during scraping.
NFR-03	Performance	PDF report generation SHALL complete within 10 seconds for any single entity report.
NFR-04	Reliability	If a scrape fails, the system SHALL fall back to the most recent SQLite data and display a visible 'Using cached data from [date]' notice. The app SHALL NOT crash.
NFR-05	Reliability	If the exchange rate API is unavailable, the system SHALL fall back to a hardcoded approximate rate and display a warning label.
NFR-06	Usability	The application SHALL be fully navigable without any onboarding or documentation.
NFR-07	Usability	All charts SHALL include titles, axis labels, and tooltips on hover.
NFR-08	Usability	All monetary figures SHALL clearly state their currency and unit (e.g. USD, INR Cr).
NFR-09	Usability	Missing or unavailable data SHALL be shown as 'N/A' or 'Data unavailable', never as a blank cell or unhandled error.
NFR-10	Compatibility	The application SHALL render correctly on Chrome, Firefox, and Safari (latest two major versions).
NFR-11	Compatibility	The application SHALL be usable on screen widths of 1024px and above (desktop / tablet landscape).
NFR-12	Scalability	The SQLite database SHALL remain performant up to 50,000 movie records and 500,000 daily performance rows.
NFR-13	Maintainability	Each scraper module SHALL be independently replaceable without modifying the business logic or UI layers.
NFR-14	Maintainability	All colour and style constants SHALL be centralised in a single theme.py file.
NFR-15	Legal	The system SHALL enforce a minimum 2-second delay between consecutive HTTP requests to the same domain.

19 UI/UX Design Specification

19.1 Dark Mode Design (Primary)

Dark mode is the default and primary design target. It should feel premium, sleek, and modern — not a generic Streamlit default.

Background	#0F172A (deep navy-black) for the main canvas. #1E293B for cards and panels.
Surface / Cards	#1E293B with a subtle 1px border at #334155. Slight border-radius (8px equivalent) for a modern card feel.
Primary Accent	#3B82F6 (blue) for CTAs, active nav items, chart highlights, and links.
Secondary Accent	#8B5CF6 (purple) for series/franchise elements. #10B981 (green) for positive verdicts. #EF4444 (red) for negative verdicts.
Text — Primary	#F1F5F9 (near-white) for headings and key figures.
Text — Secondary	#94A3B8 for labels, captions, and metadata.
Charts	plotly_dark template as base. Accent colours applied as consistent colour sequences.
Sidebar	#1E293B background. Active tab highlighted with #3B82F6 left border and slightly lighter background.

19.2 Light Mode Design

Light mode must be colourful, modern, and sleek — not a plain white page.

Background	#F8FAFC (very light grey-white). Cards use #FFFFFF with a soft box shadow.
Surface / Cards	#FFFFFF with a 1px #E2E8F0 border and subtle drop shadow. Border-radius consistent with dark mode.
Primary Accent	#2563EB (vibrant blue). Secondary: #7C3AED (purple), #059669 (green), #DC2626 (red).
Text — Primary	#0F172A for headings and key figures.
Text — Secondary	#64748B for labels and metadata.
Charts	Custom light template with white background, coloured grid lines (#E2E8F0), and the same accent colour sequence as dark mode.
Sidebar	White background with active tab highlighted by a #2563EB left border and #EFF6FF background.
Gradient Accents	Section headers and the cover area use subtle linear gradients (e.g.

	#EFF6FF to #DBEAFE) to add visual richness without being garish.
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19.3 Component Patterns

Stat Cards	Key metrics displayed as full-width or grid stat cards with a large bold number, a label below, and a thin coloured top border indicating category (blue = global, amber = India).
Film Cards	Compact cards in a grid: poster placeholder (grey box), title, year, key metric, verdict badge. Clicking navigates to detail view.
Verdict Badges	Colour-coded pill badges: Blockbuster / All-Time Blockbuster = green, Hit / Super Hit = teal, Average = amber, Flop / Disaster = red.
Charts	All charts have consistent padding, visible tooltips on hover, and a download-as-PNG icon in the Plotly toolbar.
Tables	Striped rows (alternating very subtle background). Column headers bold with background fill. Sortable columns indicated by arrow icon.
Breadcrumb	Simple text breadcrumb at top of detail views: Home > Search > [Film Name] or Home > [Series Name] > [Film Name].
Loading State	A Streamlit spinner with a contextual message (e.g. 'Fetching data from Box Office Mojo...') shown during all scrape operations.
Empty / Error	A styled empty-state card (icon + message) for no results. A red alert banner for errors, showing a human-readable message and a 'Use cached data' fallback option.

20 Scraper Contract & Output Schema

Each scraper module must conform to a defined contract. This ensures the business logic layer can consume any scraper's output without modification if a scraper is replaced.

20.1 Contract Rules

- Each scraper exposes a single public function: `scrape(entity_key: str) -> pd.DataFrame`.
- The returned DataFrame always contains the defined columns for that source, even if values are NaN for missing fields.
- All string fields are stripped, lowercased, and normalized before returning.
- All monetary fields are returned as Python float (not string). Currency symbols are never included in data fields.
- All date fields are returned as Python date objects (not strings).
- If a scrape fails for any reason, the scraper raises a `ScraperException` with a human-readable message. It never returns partial data silently.

20.2 BOM Scraper Output Schema

title_normalized	str — lowercase, stripped
title_display	str — original casing
release_date	date
worldwide_gross_usd	float — USD
domestic_gross_usd	float — USD
foreign_gross_usd	float — USD
opening_weekend_usd	float — USD, nullable
theater_count	int, nullable
days_in_release	int

20.3 Sacnilk Scraper Output Schema

title_normalized	str
title_display	str
release_date	date
language	str
india_net_cr	float — INR Crores
india_gross_cr	float — INR Crores, nullable
verdict	str
days_in_release	int

21 Error Handling & Graceful Degradation

Failure Scenario	User-Visible Behaviour	Technical Handling
Scrape fails (network error / site structure change)	Red alert banner: 'Could not fetch fresh data. Showing last available data from [date].' App continues normally using SQLite cache.	ScraperException caught at the data layer. Falls back to most recent SQLite row(s). Error logged to console.
No cached data and scrape fails	Empty-state card: 'No data available for this title yet. Try again later.'	Returns empty DataFrame. UI renders empty state component, not an error page.
Exchange rate API unavailable	Warning label on all converted figures: 'Using approximate rates. Live rates unavailable.'	Falls back to hardcoded approximate rates stored in constants.py.

Failure Scenario	User-Visible Behaviour	Technical Handling
Partial data (one source missing)	Data shown from available source. A small info badge: 'India data unavailable' or 'Global data unavailable'.	TitleMatcher returns a partial match object. UI renders available fields and marks missing fields as N/A.
PDF export fails	Error toast: 'PDF generation failed. Please try again.'	Exception caught in PDFReportBuilder. UI shows toast notification. No crash.
SQLite locked (concurrent write)	Transparent to user — slight delay.	WAL mode enabled. Retry logic (3 attempts, 200ms backoff) before raising an error.
TV data unavailable for a series	Fields shown as 'Data unavailable'. Charts that require missing data are hidden with a notice.	TVMaze returns empty list for unknown series. UI handles zero-length DataFrames in all chart components.

22 Unit Testing Strategy

CineStats uses Python's built-in unittest framework for all unit testing. Tests are organised by module and run via a single test runner. No integration or end-to-end tests are required for the initial release.

22.1 Test Framework & Tooling

- Framework: unittest (Python stdlib) — no external test runner required.
- Test files: Located in tests/ directory, mirroring the src/ structure. e.g. tests/test_bom_scraper.py for src/bom_scraper.py.
- Mocking: unittest.mock.patch and MagicMock used to mock HTTP calls and SQLite writes in all scraper and database tests.
- Test data: Static HTML fixture files stored in tests/fixtures/ for scraper parsing tests. Static CSV fixtures for matcher and calculator tests.
- Running: python -m unittest discover tests/

22.2 Test Coverage by Module

Module	Test File	Key Test Cases
bom_scraper.py	test_bom_scraper.py	Parses valid HTML fixture correctly; raises ScraperException on malformed HTML; returns correct column types; strips currency symbols from gross fields.
sacnilk_scraper.py	test_sacnilk_scraper.py	Parses India Net and verdict correctly; handles

Module	Test File	Key Test Cases
		missing verdict gracefully; normalizes title strings; raises ScraperException on HTTP error.
title_matcher.py	test_title_matcher.py	Exact match returns confidence 1.0; fuzzy match within threshold succeeds; below-threshold match returns no match; release date mismatch beyond window returns no match.
franchise_grouper.py	test_franchise_grouper.py	Films with matching normalized prefix are grouped correctly; override table entries override auto-grouping; ungrouped films return franchise_id = null.
predictor_engine.py	test_predictor_engine.py	Projection within expected range for known historical inputs; handles zero opening weekend gracefully; returns confidence interval not just point estimate.
currency_converter.py	test_currency_converter.py	Correct conversion for USD-to-INR; identity conversion when selected currency equals source currency; fallback rate used when live rate absent.
pdf_report_builder.py	test_pdf_report_builder.py	Output is a valid PDF byte stream; PDF contains expected section titles; handles empty DataFrame without exception; chart embed does not raise when kaleido is available.
init_db.py	test_init_db.py	All expected tables exist after init; upsert does not duplicate rows; schema version is recorded correctly.
mover_calculator.py	test_mover_calculator.py	Top gainer identified correctly from fixture data; percentage change calculation is accurate; handles ties deterministically.
clash_detector.py	test_clash_detector.py	Films on same date are grouped; films on different dates are not grouped; single-film date returns a group of one.

22.3 Minimum Coverage Target

- All scraper parsing functions: 100% line coverage.
- All business logic modules (matcher, grouper, predictor, calculator): 90% line coverage.
- Database layer (init, upsert, cache): 85% line coverage.
- PDF builder and currency converter: 80% line coverage.
- Streamlit UI components are excluded from unit testing; they are verified by manual review.

23 Build Plan & Milestones

Phase	Milestone	Deliverables
1	Project Setup & DB	Repository structure, requirements.txt, init_db.py with full schema, test skeleton, theme.py with dark/light palette constants.
2	Film Scrapers	bom_scraper.py and sacnilk_scraper.py with full unit tests. ScraperException and scrape_cache integration. Fixture files for tests.
3	TV & Exchange Rate	tmvaze_client.py, wikipedia_tv_scraper.py with unit tests. currency_converter.py with fallback logic and unit tests.
4	Business Logic	title_matcher.py, franchise_grouper.py, predictor_engine.py, clash_detector.py, mover_calculator.py — all with unit tests.
5	Home Tab	Streamlit app.py + home.py with Top of Year, Releases Today/This Week, and Now Playing sections. Dark mode. Plotly charts. Session cache.
6	Detail Views	movie_detail.py, series_franchise.py, tv_series_detail.py. All charts, stat cards, regional breakdown, On This Day.
7	Search & Sidebar	search.py with filters, result cards, recent searches. Sidebar navigation. Breadcrumb. Light mode toggle functional.
8	Compare Tab	compare.py: Film comparison (up to 10) and TV comparison (up to 10, separate). All charts and metrics tables.
9	PDF Export	pdf_report_builder.py with kaleido chart export. PDF button on all detail and compare views. Unit tests.
10	Currency & Settings	Settings panel. Currency selector end-to-end. Bracket display logic for worldwide gross. Rate staleness label.
11	Polish, Error Handling, QA	Empty states, error banners, loading spinners, fallback flows for all failure scenarios. Full test suite passing. Manual UI review.
12	Deployment	Deploy to Streamlit Community Cloud via GitHub. README with setup instructions. Final requirements.txt freeze.

24 Configuration & Environment Setup

24.1 Repository Structure

cinestats/	Root of the repository.
app.py	Streamlit entry point.
requirements.txt	Pinned dependency list.
cinestats.db	SQLite database file (gitignored).
src/	All Python source modules: scrapers, matchers, calculators, converters, pdf builder.

pages/	One Streamlit page module per tab: home.py, search.py, compare.py, tv_series.py, settings.py.
components/	Reusable Streamlit UI components.
theme.py	Centralised colour and style constants.
tests/	All unittest test files, mirroring src/ structure.
tests/fixtures/	Static HTML and CSV fixtures for unit tests.
.streamlit/ config.toml	Streamlit config: theme base, server settings.

24.2 Key Dependencies (requirements.txt)

streamlit	UI framework.
pandas	Data processing.
requests	HTTP scraping.
beautifulsoup4	HTML parsing.
lxml	HTML/XML parser backend for BeautifulSoup and pandas.read_html.
rapidfuzz	Fuzzy string matching for title matcher.
plotly	Interactive charts.
kaleido	Static PNG export of Plotly charts for PDF embedding.
reportlab	PDF generation.
tvmaze (or direct requests to TVMaze REST API)	TV series data.

24.3 Streamlit Configuration (.streamlit/config.toml)

[theme] base	"dark" — sets Streamlit's base theme to dark.
primaryColor	#3B82F6 — primary accent blue.
backgroundColor	#0F172A — main canvas background.
secondaryBackgroundColor or	#1E293B — sidebar and card background.
textColor	#F1F5F9 — primary text.
[server] maxUploadSize	Not applicable (no user uploads). Set to default.

24.4 Local Development Setup

8. Clone the repository: `git clone https://github.com/[user]/cinestats.git`
9. Create a virtual environment: `python -m venv .venv && source .venv/bin/activate`
10. Install dependencies: `pip install -r requirements.txt`
11. Initialise the database: `python src/init_db.py`
12. Run the app locally: `streamlit run app.py`
13. Run unit tests: `python -m unittest discover tests/`

25 Legal & Ethical Considerations

- Personal and Educational Use Only: CineStats is not intended for commercial redistribution. All scraped data is used solely for display and analysis within the application.
- Robots.txt Compliance: All scrapers must check and respect the robots.txt of each source domain before initiating requests.
- Rate Limiting: A minimum delay of 2 seconds between consecutive requests to the same domain is enforced in all scrapers. Concurrent scraping of the same domain is not permitted.
- User-Agent Headers: All HTTP requests must include an honest, descriptive User-Agent string: CineStats-Bot/2.0 (personal project; non-commercial).
- No Login Circumvention: Scrapers must not bypass authentication, CAPTCHA, paywalls, or any access control mechanism.
- Session Cache: The session-based cache directly limits the frequency of scraping. The same data set is never fetched more than once per session.
- No PII Collection: CineStats does not collect, store, or transmit any personally identifiable information. The watchlist is stored locally in SQLite only.
- No Commercial Redistribution: Scraped data is used solely within CineStats. It is not sold, published, or shared via an API.
- Monetization: CineStats is free forever with no advertising, no sponsored content, and no data monetisation of any kind.

26 Future Scope & Roadmap

26.1 Near-Term (v2.1 – v2.3)

- TMDb API Integration: Pull movie posters, genre tags, cast lists, and trailer links to enrich Movie Detail views.
- Automated Daily Data Refresh: GitHub Actions scheduled job to update the SQLite database daily and push changes to the deployed app.
- Milestone Alerts (Console): Display a milestone achievement banner on the detail view when a film crosses a configurable threshold (e.g. crossed ₹100 Cr India Net today).

26.2 Medium-Term (v3.0)

- OTT Tracking: Track when films move to streaming and display streaming release dates alongside theatrical data.
- Predictive Analytics: ML-based gross trajectory prediction using opening weekend, screen count, genre, franchise history, and seasonal factors.
- Historical Archive: Year-on-year comparison tools with a dedicated historical analysis view.
- Genre & Studio Deep-Dives: Dedicated studio performance views with year-on-year trend charts.

26.3 Long-Term (v4.0+)

- User Accounts: Persistent cross-session watchlists and personalised dashboards.
- Social Sentiment Correlation: Correlate IMDb/RT score movements with box office trajectory changes.
- Franchise Prediction: Predict opening weekend of upcoming sequels based on franchise trajectory and external factors.
- Mobile-Optimised Layout: Responsive design extending support to mobile screen widths.

27 Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Source site HTML structure changes	High	Scrapers break, no fresh data.	Modular scrapers. Schema validation on output. Unit tests with fixtures catch structure changes early. Fallback to SQLite cache.
IP rate-limiting or blocking	Medium	Scrape failures, data gaps.	Minimum 2s delays. Session cache limits frequency. Exponential backoff on retries. Graceful fallback to cached data.
TVMaze API changes or deprecation	Low	TV module data gaps.	TVMaze is a stable free API. Direct scraping of Wikipedia TV tables as fallback.
kaleido chart export failures in Streamlit Cloud	Medium	PDF charts missing or blank.	Unit test PDF builder. Pin kaleido version. Detect failure and generate PDF without charts rather than crashing.
SQLite concurrency (multiple sessions)	Medium	Potential write contention.	WAL journal mode. Retry logic (3x, 200ms backoff). Read-only fallback if write lock cannot be acquired.
Title matching errors	Medium	Incorrect data associations shown.	Confidence score stored and displayed. Manual override table for known edge cases. Flagged matches shown with a warning badge in UI.
Streamlit Community Cloud cold start	Medium	Slow first load for new	Pre-populate SQLite on deploy. Show loading spinner with contextual message. Home tab

Risk	Likelihood	Impact	Mitigation
latency		sessions.	prioritises cached data.
Exchange rate API unavailability	Low	Incorrect or absent currency conversion.	Hardcoded approximate fallback rates. Visible warning label. No feature blocked by this failure.

28 Glossary

India Net	Box office collections in India after deducting GST and entertainment tax. The standard industry metric for Indian films.
India Gross	Total box office collections in India before tax deductions.
Worldwide Gross	Total global box office revenue in USD across all territories.
Domestic (US)	Box office collections specifically within the United States and Canada.
Verdict	Industry shorthand for commercial performance vs budget: All-Time Blockbuster, Blockbuster, Super Hit, Hit, Average, Flop, Disaster.
Now Playing	Films currently running in cinemas — i.e., not yet finished their theatrical run.
Clash	Two or more films sharing the same theatrical release date and competing for audience share.
Holdover	A film's week-on-week retention of audience attendance. High holdover indicates strong word-of-mouth.
BOM	Box Office Mojo — primary global box office data aggregator, owned by IMDb/Amazon.
Sacnilk	Leading Indian box office tracking website covering Bollywood and all major regional industries.
TVMaze	A free REST API providing structured TV series, episode, and network data.
Upsert	Database operation: INSERT if the record does not exist, UPDATE if it does. Uses INSERT OR REPLACE in SQLite.
WAL Mode	Write-Ahead Logging — SQLite journaling mode that improves concurrent read/write performance.
Session Cache	A record in SQLite's scrape_cache table tracking what data has been fetched in the current Streamlit session, preventing re-scraping.
Franchise	A collection of films sharing a common cinematic universe, series, or brand (e.g. Marvel Cinematic Universe, Fast & Furious).
rapidfuzz	Python library for fast fuzzy string matching, used in the title matcher to align film names across sources.

kaleido	Python library that enables static image export of Plotly charts, used to embed charts in PDF reports.
ReportLab	Python library for programmatic PDF generation, used to build all CineStats PDF report exports.

End of Document — CineStats v2.0 Project Specification